

[INICIAL] TCArg 2026 Contest #6

A. Timofey and a tree

2 seconds, 256 megabytes

Each New Year Timofey and his friends cut down a tree of n vertices and bring it home. After that they paint all the n its vertices, so that the i -th vertex gets color c_i .

Now it's time for Timofey birthday, and his mother asked him to remove the tree. Timofey removes the tree in the following way: he takes some vertex in hands, while all the other vertices move down so that the tree becomes rooted at the chosen vertex. After that Timofey brings the tree to a trash can.

Timofey doesn't like it when many colors are mixing together. A subtree annoys him if there are vertices of different color in it. Timofey wants to find a vertex which he should take in hands so that there are no subtrees that annoy him. He doesn't consider the whole tree as a subtree since he can't see the color of the root vertex.

A subtree of some vertex is a subgraph containing that vertex and all its descendants.

Your task is to determine if there is a vertex, taking which in hands Timofey wouldn't be annoyed.

Input

The first line contains single integer n ($2 \leq n \leq 10^5$) — the number of vertices in the tree.

Each of the next $n - 1$ lines contains two integers u and v ($1 \leq u, v \leq n, u \neq v$), denoting there is an edge between vertices u and v . It is guaranteed that the given graph is a tree.

The next line contains n integers $c_1, c_2, \dots, c_n$ ($1 \leq c_i \leq 10^5$), denoting the colors of the vertices.

Output

Print "NO" in a single line, if Timofey can't take the tree in such a way that it doesn't annoy him.

Otherwise print "YES" in the first line. In the second line print the index of the vertex which Timofey should take in hands. If there are multiple answers, print any of them.

input
4 1 2 2 3 3 4 1 2 1 1
output
YES 2

input
3 1 2 2 3 1 2 3
output
YES 2

input
4 1 2 2 3 3 4 1 2 1 2
output
NO

B. Gifts Fixing

1 second, 256 megabytes

You have n gifts and you want to give all of them to children. Of course, you don't want to offend anyone, so all gifts should be equal between each other. The i -th gift consists of a_i candies and b_i oranges.

During one move, you can choose some gift $1 \leq i \leq n$ and do one of the following operations:

- eat exactly **one candy** from this gift (decrease a_i by one);
- eat exactly **one orange** from this gift (decrease b_i by one);

- eat exactly **one candy** and exactly **one orange** from this gift (decrease both a_i and b_i by one).

Of course, you can not eat a candy or orange if it's not present in the gift (so neither a_i nor b_i can become less than zero).

As said above, all gifts should be equal. This means that after some sequence of moves the following two conditions should be satisfied: $a_1 = a_2 = \dots = a_n$ and $b_1 = b_2 = \dots = b_n$ (and a_i equals b_i is **not necessary**).

Your task is to find the **minimum** number of moves required to equalize all the given gifts.

You have to answer t independent test cases.

Input

The first line of the input contains one integer t ($1 \leq t \leq 1000$) — the number of test cases. Then t test cases follow.

The first line of the test case contains one integer n ($1 \leq n \leq 50$) — the number of gifts. The second line of the test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$), where a_i is the number of candies in the i -th gift. The third line of the test case contains n integers $b_1, b_2, \dots, b_n$ ($1 \leq b_i \leq 10^9$), where b_i is the number of oranges in the i -th gift.

Output

For each test case, print one integer: the **minimum** number of moves required to equalize all the given gifts.

input
5 3 3 5 6 3 2 3 5 1 2 3 4 5 5 4 3 2 1 3 1 1 1 2 2 2 6 1 1000000000 1000000000 1000000000 1000000000 1000000000 1 1 1 1 1 1 3 10 12 8 7 5 4
output
6 16 0 4999999995 7

In the first test case of the example, we can perform the following sequence of moves:

- choose the first gift and eat one orange from it, so $a = [3, 5, 6]$ and $b = [2, 2, 3]$;
- choose the second gift and eat one candy from it, so $a = [3, 4, 6]$ and $b = [2, 2, 3]$;
- choose the second gift and eat one candy from it, so $a = [3, 3, 6]$ and $b = [2, 2, 3]$;
- choose the third gift and eat one candy and one orange from it, so $a = [3, 3, 5]$ and $b = [2, 2, 2]$;
- choose the third gift and eat one candy from it, so $a = [3, 3, 4]$ and $b = [2, 2, 2]$;
- choose the third gift and eat one candy from it, so $a = [3, 3, 3]$ and $b = [2, 2, 2]$.

C. TMT Document

1 second, 256 megabytes

The student council has a shared document file. Every day, some members of the student council write the sequence TMT (short for Towa Maji Tenshi) in it.

However, one day, the members somehow entered the sequence into the document at the same time, creating a jumbled mess. Therefore, it is Suguru Doujima's task to figure out whether the document has malfunctioned. Specifically, he is given a string of length n whose characters are all either T or M, and he wants to figure out if it is possible to partition it into some number of disjoint subsequences, all of which are equal to TMT. That is, each character of the string should belong to exactly one of the subsequences.

A string a is a subsequence of a string b if a can be obtained from b by deletion of several (possibly, zero) characters.

Input

The first line contains an integer t ($1 \leq t \leq 5000$) — the number of test cases.

The first line of each test case contains an integer n ($3 \leq n < 10^5$), the number of characters in the string entered in the document. It is guaranteed that n is divisible by 3.

The second line of each test case contains a string of length n consisting of only the characters T and M.

It is guaranteed that the sum of n over all test cases does not exceed 10^5 .

Output

For each test case, print a single line containing YES if the described partition exists, and a single line containing NO otherwise.

input
5 3 TMT 3 MTT 6 TMTMTT 6 TMTTTT 6 TTMMTT
output
YES NO YES NO YES

In the first test case, the string itself is already a sequence equal to TMT.

In the third test case, we may partition the string into the subsequences **TMT****MTT**. Both the bolded and the non-bolded subsequences are equal to TMT.

D. Inverse of Rows and Columns

2 seconds, 256 megabytes

You are given a binary matrix a of size $n \times m$. A binary matrix is a matrix where each element is either 0 or 1.

You may perform some (possibly zero) operations with this matrix. During each operation you can inverse the row of this matrix or a column of this matrix. Formally, inverting a row is changing all values in this row to the opposite (0 to 1, 1 to 0). Inverting a column is changing all values in this column to the opposite.

Your task is to sort the initial matrix by some sequence of such operations. The matrix is considered **sorted** if the array $[a_{1,1}, a_{1,2}, \dots, a_{1,m}, a_{2,1}, a_{2,2}, \dots, a_{2,m}, \dots, a_{n,m-1}, a_{n,m}]$ is sorted in non-descending order.

Input

The first line of the input contains two integers n and m ($1 \leq n, m \leq 200$) — the number of rows and the number of columns in the matrix.

The next n lines contain m integers each. The j -th element in the i -th line is $a_{i,j}$ ($0 \leq a_{i,j} \leq 1$) — the element of a at position (i, j) .

Output

If it is impossible to obtain a **sorted** matrix, print "NO" in the first line.

Otherwise print "YES" in the first line. In the second line print a string r of length n . The i -th character r_i of this string should be '1' if the i -th row of the matrix is inverted and '0' otherwise. In the third line print a string c of length m . The j -th character c_j of this string should be '1' if the j -th column of the matrix is inverted and '0' otherwise. If there are multiple answers, you can print any.

input
2 2 1 1 0 1
output
YES 00 10

input
3 4 0 0 0 1 0 0 0 0 1 1 1 1
output
YES 010 0000

input
3 3 0 0 0 1 0 1 1 1 0

output
NO

E. Close Tuples

2 s., 256 MB

You are given a sequence a of length n consisting of integers from 1 to n . **The sequence may contain duplicates (i.e. some elements can be equal).**

Find the number of tuples of $m = 3$ elements such that the maximum number in the tuple differs from the minimum by no more than $k = 2$. Formally, you need to find the number of triples of indices $i < j < z$ such that

$$\max(a_i, a_j, a_z) - \min(a_i, a_j, a_z) \leq 2.$$

For example, if $n = 4$ and $a = [1, 2, 4, 3]$, then there are two such triples ($i = 1, j = 2, z = 4$ and $i = 2, j = 3, z = 4$). If $n = 4$ and $a = [1, 1, 1, 1]$, then all four possible triples are suitable.

Input

The first line contains a single integer t ($1 \leq t \leq 2 \cdot 10^5$) — the number of test cases. Then t test cases follow.

The first line of each test case contains an integer n ($1 \leq n \leq 2 \cdot 10^5$) — the length of the sequence a .

The next line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$) — the sequence a .

It is guaranteed that the sum of n for all test cases does not exceed $2 \cdot 10^5$.

Output

Output t answers to the given test cases. Each answer is the required number of triples of elements, such that the maximum value in the triple differs from the minimum by no more than 2. Note that in difference to the hard version of the problem, you **don't need** to output the answer by modulo. You must output the exact value of the answer.

input
4 4 1 2 4 3 4 1 1 1 1 1 1 10 5 6 1 3 2 9 8 1 2 4
output
2 4 0 15

F. Salazar Slytherin's Locket

2 seconds, 256 megabytes

Harry came to know from Dumbledore that Salazar Slytherin's locket is a horcrux. This locket was present earlier at 12 Grimmauld Place, the home of Sirius Black's mother. It was stolen from there and is now present in the Ministry of Magic in the office of Dolorous Umbridge, Harry's former Defense Against the Dark Arts teacher.

Harry, Ron and Hermione are infiltrating the Ministry. Upon reaching Umbridge's office, they observed a code lock with a puzzle asking them to calculate count of magic numbers between two integers l and r (both inclusive).

Harry remembered from his detention time with Umbridge that she defined a magic number as a number which when converted to a given base b , all the digits from 0 to $b - 1$ appear even number of times in its representation without any leading zeros.

You have to answer q queries to unlock the office. Each query has three integers b_i, l_i and r_i , the base and the range for which you have to find the count of magic numbers.

Input

First line of input contains q ($1 \leq q \leq 10^5$) — number of queries.

Each of the next q lines contain three space separated integers b_i, l_i, r_i ($2 \leq b_i \leq 10, 1 \leq l_i \leq r_i \leq 10^{18}$).

Output

You have to output q lines, each containing a single integer, the answer to the corresponding query.

input
2 2 4 9 3 1 10

output
1 2

input
2 2 1 100 5 1 100
output
21 4

In sample test case 1, for first query, when we convert numbers 4 to 9 into base 2, we get:

- $4 = 100_2$,
- $5 = 101_2$,
- $6 = 110_2$,
- $7 = 111_2$,
- $8 = 1000_2$,
- $9 = 1001_2$.

Out of these, only base 2 representation of 9 has even number of 1 and 0. Thus, the answer is 1.

G. Same Parity Summands

1 second, 256 megabytes

You are given two positive integers n ($1 \leq n \leq 10^9$) and k ($1 \leq k \leq 100$). Represent the number n as the sum of k positive integers of the same parity (have the same remainder when divided by 2).

In other words, find $a_1, a_2, \dots, a_k$ such that all $a_i > 0$, $n = a_1 + a_2 + \dots + a_k$ and either all a_i are even or all a_i are odd at the same time.

If such a representation does not exist, then report it.

Input

The first line contains an integer t ($1 \leq t \leq 1000$) — the number of test cases in the input. Next, t test cases are given, one per line.

Each test case is two positive integers n ($1 \leq n \leq 10^9$) and k ($1 \leq k \leq 100$).

Output

For each test case print:

- YES and the required values a_i , if the answer exists (if there are several answers, print any of them);
- NO if the answer does not exist.

The letters in the words YES and NO can be printed in any case.

input
8 10 3 100 4 8 7 97 2 8 8 3 10 5 3 1000000000 9
output
YES 4 2 4 YES 55 5 5 35 NO NO YES 1 1 1 1 1 1 1 1 NO YES 3 1 1 YES 111111110 111111110 111111110 111111110 111111110 111111110 111111110 111111110 111111120

H. Cut Ribbon

1 second, 256 megabytes

Polycarpus has a ribbon, its length is n . He wants to cut the ribbon in a way that fulfils the following two conditions:

- After the cutting each ribbon piece should have length a , b or c .
- After the cutting the number of ribbon pieces should be maximum.

Help Polycarpus and find the number of ribbon pieces after the required cutting.

Input

The first line contains four space-separated integers n , a , b and c ($1 \leq n, a, b, c \leq 4000$) — the length of the original ribbon and the acceptable lengths of the ribbon pieces after the cutting, correspondingly. The numbers a , b and c can coincide.

Output

Print a single number — the maximum possible number of ribbon pieces. It is guaranteed that at least one correct ribbon cutting exists.

input
5 5 3 2
output
2

input
7 5 5 2
output
2

In the first example Polycarpus can cut the ribbon in such way: the first piece has length 2, the second piece has length 3.

In the second example Polycarpus can cut the ribbon in such way: the first piece has length 5, the second piece has length 2.

I. The Number of Subpermutations

2 seconds, 256 megabytes

You have an array $a_1, a_2, \dots, a_n$.

Let's call some subarray $a_l, a_{l+1}, \dots, a_r$ of this array a *subpermutation* if it contains all integers from 1 to $r - l + 1$ exactly once. For example, array $a = [2, 2, 1, 3, 2, 3, 1]$ contains 6 subarrays which are subpermutations: $[a_2 \dots a_3]$, $[a_2 \dots a_4]$, $[a_3 \dots a_3]$, $[a_3 \dots a_5]$, $[a_5 \dots a_7]$, $[a_7 \dots a_7]$.

You are asked to calculate the number of subpermutations.

Input

The first line contains one integer n ($1 \leq n \leq 3 \cdot 10^5$).
The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq n$).

This array can contain the same integers.

Output

Print the number of subpermutations of the array a .

input
8 2 4 1 3 4 2 1 2
output
7

input
5 1 1 2 1 2
output
6

There are 7 subpermutations in the first test case. Their segments of indices are $[1, 4]$, $[3, 3]$, $[3, 6]$, $[4, 7]$, $[6, 7]$, $[7, 7]$ and $[7, 8]$.

In the second test case 6 subpermutations exist: $[1, 1]$, $[2, 2]$, $[2, 3]$, $[3, 4]$, $[4, 4]$ and $[4, 5]$.

J. Ordinary Numbers

2 seconds, 256 megabytes

Let's call a positive integer n ordinary if in the decimal notation all its digits are the same. For example, 1, 2 and 99 are ordinary numbers, but 719 and 2021 are not ordinary numbers.

For a given number n , find the number of ordinary numbers among the numbers from 1 to n .

Input

The first line contains one integer t ($1 \leq t \leq 10^4$). Then t test cases follow.

Each test case is characterized by one integer n ($1 \leq n \leq 10^9$).

Output

For each test case output the number of ordinary numbers among numbers from 1 to n .

input
6 1 2 3 4 5 100
output
1 2 3 4 5 18

K. Harmonious Graph

1 second, 256 megabytes

You're given an undirected graph with n nodes and m edges. Nodes are numbered from 1 to n .

The graph is considered *harmonious* if and only if the following property holds:

- For every triple of integers (l, m, r) such that $1 \leq l < m < r \leq n$, if there exists a **path** going from node l to node r , then there exists a **path** going from node l to node m .

In other words, in a harmonious graph, if from a node l we can reach a node r through edges $(l < r)$, then we should able to reach nodes $(l + 1), (l + 2), \dots, (r - 1)$ too.

What is the minimum number of edges we need to add to make the graph harmonious?

Input

The first line contains two integers n and m ($3 \leq n \leq 200\,000$ and $1 \leq m \leq 200\,000$).

The i -th of the next m lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n, u_i \neq v_i$), that mean that there's an edge between nodes u and v .

It is guaranteed that the given graph is simple (there is no self-loop, and there is at most one edge between every pair of nodes).

Output

Print the minimum number of edges we have to add to the graph to make it harmonious.

input
14 8 1 2 2 7 3 4 6 3 5 7 3 8 6 8 11 12
output
1

input
200000 3 7 9 9 8 4 5
output
0

In the first example, the given graph is not harmonious (for instance, $1 < 6 < 7$, node 1 can reach node 7 through the path $1 \rightarrow 2 \rightarrow 7$, but node 1 can't reach node 6). However adding the edge $(2, 4)$ is sufficient to make it harmonious.

In the second example, the given graph is already harmonious.

L. Road Reform

2 seconds, 512 megabytes

There are n cities and m bidirectional roads in Berland. The i -th road connects the cities x_i and y_i , and has the speed limit s_i . The road network allows everyone to get from any city to any other city.

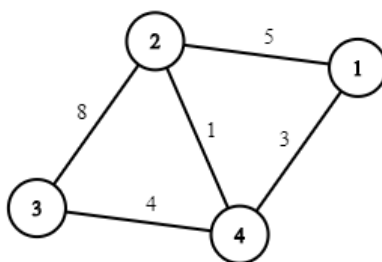
The Berland Transport Ministry is planning a road reform.

First of all, maintaining all m roads is too costly, so $m - (n - 1)$ roads will be demolished in such a way that the remaining $(n - 1)$ roads still allow to get to any city from any other city. Formally, the remaining roads should represent an undirected tree.

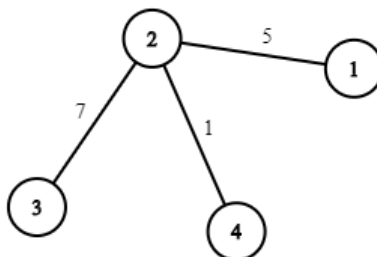
Secondly, the speed limits on the remaining roads might be changed. The changes will be done sequentially, each change is either increasing the speed limit on some road by 1, or decreasing it by 1. Since changing the speed limit requires a lot of work, the Ministry wants to minimize the number of changes.

The goal of the Ministry is to have a road network of $(n - 1)$ roads with the maximum speed limit over all roads equal to exactly k . They assigned you the task of calculating the minimum number of speed limit changes they have to perform so the road network meets their requirements.

For example, suppose the initial map of Berland looks like that, and $k = 7$:



Then one of the optimal courses of action is to demolish the roads 1–4 and 3–4, and then decrease the speed limit on the road 2–3 by 1, so the resulting road network looks like that:

**Input**

The first line contains one integer t ($1 \leq t \leq 1000$) — the number of test cases.

The first line of each test case contains three integers n , m and k ($2 \leq n \leq 2 \cdot 10^5$; $n - 1 \leq m \leq \min(2 \cdot 10^5, \frac{n(n-1)}{2})$; $1 \leq k \leq 10^9$) — the number of cities, the number of roads and the required maximum speed limit, respectively.

Then m lines follow. The i -th line contains three integers x_i , y_i and s_i ($1 \leq x_i, y_i \leq n$; $x_i \neq y_i$; $1 \leq s_i \leq 10^9$) — the cities connected by the i -th road and the speed limit on it, respectively. All roads are bidirectional.

The road network in each test case is connected (that is, it is possible to reach any city from any other city by traveling along the road), and each pair of cities is connected by at most one road.

The sum of n over all test cases does not exceed $2 \cdot 10^5$. Similarly, the sum of m over all test cases does not exceed $2 \cdot 10^5$.

Output

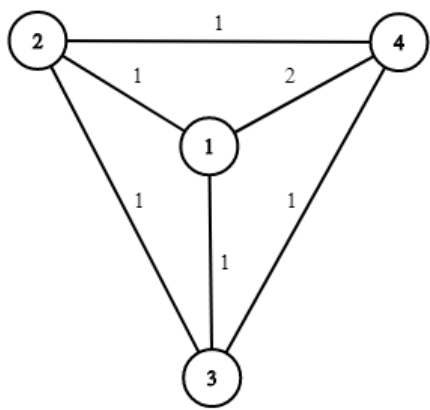
For each test case, print one integer — the minimum number of changes the Ministry has to perform so that the maximum speed limit among the remaining $(n - 1)$ roads is exactly k .

input
4 4 5 7 4 1 3 1 2 5 2 3 8 2 4 1 3 4 4 4 6 5 1 2 1 1 3 1 1 4 2 2 4 1 4 3 1 3 2 1 3 2 10 1 2 8 1 3 10 5 5 15 1 2 17 3 1 15 2 3 10 1 4 14 2 5 8
output
1 3 0 0

The explanation for the example test:

The first test case is described in the problem statement.

In the second test case, the road network initially looks like that:



The Ministry can demolish the roads 1–2, 3–2 and 3–4, and then increase the speed limit on the road 1–4 three times.

In the third test case, the road network already meets all the requirements.

In the fourth test case, it is enough to demolish the road 1–2 so the resulting road network meets the requirements.

M. K-th Not Divisible by n

1 second, 256 megabytes

You are given two positive integers n and k . Print the k -th positive integer that is not divisible by n .

For example, if $n = 3$, and $k = 7$, then all numbers that are not divisible by 3 are: 1, 2, 4, 5, 7, 8, 10, 11, 13 The 7-th number among them is 10.

Input

The first line contains an integer t ($1 \leq t \leq 1000$) — the number of test cases in the input. Next, t test cases are given, one per line.

Each test case is two positive integers n ($2 \leq n \leq 10^9$) and k ($1 \leq k \leq 10^9$).

Output

For each test case print the k -th positive integer that is not divisible by n .

input

```
6
3 7
4 12
2 1000000000
7 97
1000000000 1000000000
2 1
```

output

```
10
15
1999999999
113
1000000001
1
```

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